Designing for Speech Practice Systems: How Do User-Controlled Voice Manipulation and Model Speakers Impact Self-Perceptions of Voice?

Lisa Orii
lisaorii@cs.washington.edu
Paul G. Allen School of Computer Science & Engineering,
University of Washington
Seattle, WA, USA

Nami Ogawa
ogawa@cyber.t.u-tokyo.ac.jp
The University of Tokyo
Tokyo, Japan

Yuji Hatada
hatada@cyber.t.u-tokyo.ac.jp
The University of Tokyo
Tokyo, Japan

Takuji Narumi
narumi@cyber.t.u-tokyo.ac.jp
The University of Tokyo
Tokyo, Japan

ABSTRACT

Can you speak the way you desire without feeling the pressure to conform to standards of speaking? In this study, we investigated the impact of user-controlled voice manipulation and listening to recordings of model speakers on self-perceptions of voice and speech. Quantitative analysis showed that there was a significant improvement in the perceived confidence of tone by listening to model speakers, but there were no significant improvements due to voice manipulation. Qualitative analysis of interviews revealed that participants responded positively to the visual and auditory feedback provided by the voice manipulation software. The participants also evaluated the quality of model speakers to decide whether or not they wanted to refer to them for speech practice. Based on the results of these analyses, we summarized the design implications for a speech practice system that would allow further investigation of the impact of the system on self-perceptions of speech performance.

CCS CONCEPTS

• **Human-centered computing** → **Auditory feedback**; **Empirical studies in HCI**.

KEYWORDS

public speaking, presentation training, voice manipulation, speech examples, self-perception

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1 INTRODUCTION

Speaking is critical to human communication, but identifying concrete approaches to improve speech performance may be difficult. To address this problem, researchers in the HCI community have proposed speech practice systems that provide feedback on ways to improve speech quality [13, 22, 41, 50, 53], and applications of altered auditory feedback (AAF) to alleviate speaking tensions [30]. While these speech practice systems have been found to be effective for improving speech performance, they are limited to providing instructions for improvement and evaluations of the user's speech. In other words, existing speech practice systems restrict opportunities for the user to self-assess their speech performance and determine their ideal voice.

Reflections on self-perceptions can increase the self-awareness [10, 37, 57] and the self-perception of communication skills [11, 29], improve the process of skill acquisition [15], and enhance self-confidence and self-esteem [28, 32, 49]. Especially for women and gender minorities whose confidence impacts their perceived ability to give a speech [8], self-perception is an important factor that influences speech performance. It is critical to support speakers with increasing their self-perception of voice and speech, especially because increased self-perceptions can positively impact social well-being [35]. However, there has yet to be work on a speech practice system that values the user's self-perceptions of their voice and abilities as a speaker.

In this study, we examine how a speech practice system can assist women and gender minorities with finding their desirable voices and improving their self-perceptions of voice and speech. We focus on women and gender minorities to uplift these individuals' experiences with voice and public speaking, which may be influenced by the social pressure to emulate men's "masculine" and authoritative voices [46]. We investigate the following research questions:

- **RQ1**: How does engaging with user-controlled voice manipulation influence the self-perception of voice and speech?

- **RQ2**: How does listening to recordings of model speakers influence the self-perception of voice and speech?
- **RQ3**: What design considerations should be made in a speech practice system to help prospective users with improving the self-perception of voice and speech?

We hypothesized that when users gained full control of exploring and modulating their vocal parameters, their self-perceptions of their voice would increase. We studied the impact of the self-manipulation of vocal parameters through AAF, which we refer to as "voice manipulation," on self-perceptions of voice and speech. Additionally, given prior work which has identified the positive impact of learning from role models [24, 33, 42, 45], we also examined how listening to model public speakers can influence self-perceptions of voice.

We observed the changes in self-perceptions of voice and speech from before and after the participants practiced their speeches using either, both, or none of voice manipulation and listening to recordings of example model speakers. With respect to **RQ1**, there was a trend to have higher perceived abilities to self-express through voice using voice manipulation without listening to model speakers, than when the participants did not use voice manipulation and did not listen to model speakers. With respect to **RQ2**, the results showed that only the confidence in tone significantly improved by listening to recordings of model speakers. To answer **RQ3**, we applied our findings to introduce design implications for a speech practice system that could better measure the impact of using voice manipulation and listening to recordings of model speakers on self-perceptions of voice and speech.

In sum, this paper contributes: (i) results from an experiment that assessed the impact of voice manipulation and listening to model speakers on the self-perceptions of voice and speech; (ii) an analysis of interviews with that described the participants' thought processes during the experiment and their social experiences that influenced their self-perceptions; and (iii) design recommendations for a speech practice system that could support users with improving their self-perceptions of voice and speech.

2 RELATED WORK

2.1 Self-Assessment in Public Speaking

Social persuasion, which is the judgement, feedback, and appraisal given to the speaker by listeners [54], influences the speaker's speaking skills and the self-perception of speaking. While social persuasion can impact any speaker, women, whose voice pitch is on average almost twice as high as men's voices [52], tend to face harsher social persuasion, namely the pressure to lower their voices to resemble men's voices [46]. Given that authority and leadership are more commonly associated with lower-pitched, "masculine" voices [4, 21, 46, 51], the pressure to lower vocal pitch can exclude women and gender minorities, who are less likely to have "masculine" voices, from speech-making, debate, and commentary [7]. It is worth investigating speech practice techniques that can minimize the impact of discouraging social persuasion, and value individuals' self-perceptions of their voices.

One approach that has been found to increase self-perception is self-assessment, which is the act of self-monitoring to enhance learning and performance. Self-assessments are used in public speaking practice to increase the awareness of presentation skills [10, 37, 57], improve the acquisition process [15], and increase the self-perception of communication skill achievement [11, 29]. To combat a lack of confidence that may inhibit communication [36], self-assessment practices have been used to enhance self-confidence and self-esteem [28, 32, 49]. Furthermore, self-assessing communication is effective for increasing self-expression [20], which consists of speaking as much as one wants to, saying in one's own words what one wants to say, and asking all of the questions that one wants to ask.

2.2 Techniques for Public Speaking Practice

We surveyed the literature that studied the impact of interventions in speech practice on speech performance and the self-perception of public speaking.

2.2.1 Learning from Role Models of Public Speaking. Vicarious experiences in public speaking, or the observation of a strategy to strive for similar results, can be had through learning from peers, lecturers, or exemplary videos of speech performances [33]. Prior work has found that when university students observed their peers or friends during public speaking practices, they were aware of their public speaking self-efficacy [33] and experienced an increase in public speaking self-efficacy [42]. The exposure to role models who can communicate effectively also encourages informative communication [45].

Previous research has attended to the impact of the gender of role models on learning through vicarious experiences. When women speakers were exposed to women role models, the speakers improved in their oral presentations and had more positive self-evaluations of their speech than when they were exposed to men role models or no role models [24]. These findings imply that vicarious experiences are effective when learners observe someone with similar attributes as themselves[6].

2.2.2 Speech Training Systems. Within the HCI community, researchers have developed and tested speech training systems that offer automated feedback on users' speech quality, verbal behaviors, and nonverbal behaviors during public speaking [16, 22, 50, 53]. Some speech training systems recommend recordings of example speakers so that the user can gain inspiration from examples and emulate their speaking [55]. Research on speech training has also studied AAF, which has been used to alleviate speech difficulties and speaking anxiety [47, 48] and found to impact emotions [5]. AAF has been applied to speech training systems to resynthesize voice inputs to assist users with hearing what their voices sound like with vocal modulation [39] and to generate confident-sounding versions of users' voices [30].

Existing speech training systems are motivated to help users with producing desirable public speech performances, either through incorporating instructions for users to follow or providing evaluations of users' speeches. Such external assessments and instructions limit the opportunity for users to derail from the system's instructions, and instead rely on their self-assessment and self-perceptions of their speech performance. Given the positive impact of self-assessment on self-perceptions, skill awareness, and skill acquisition, it is worth investigating how a speech practice system

that assists users with self-assessing their speeches, can support users with improving their perceptions of themselves as public speakers.

3 METHOD

The experiment was conducted in three parts: initial session, main experiment, final session. During the initial session, the participants answered a questionnaire on self-esteem, self-efficacy, and self-perceptions of voice and speech. In the main experiment, the participants practiced public speaking under four conditions, in which the participants used either, none, or both of voice manipulation and listening to recordings of model speakers (hereinafter referred to as "model audio"). Before and after each condition was tested, the participants answered a questionnaire to self-assess their voice and speech. During the final session, the participants assessed the quality of the model audio, compared the conditions they conducted in the main experiment, and self-assessed their voice and speech. Finally, semi-structured interviews were conducted with each participant.

3.1 Participants

With an interest in understanding the self-perceptions of voice and speech for those who may not have vocal traits typically observed in men, we recruited participants who identify as women or other gender minorities. Through the first author's personal networks, we recruited 17 cisgender women and 1 non-binary who were AFAB (assigned female at birth) [1] between 18 to 23 years old. One participant expressed that they were not a native English speaker. Upon recruitment, we explained the purpose and procedure of the experiment and the required commitment. Each participant was compensated with a $50 Amazon gift card. This study was approved by the ethics committee of the Graduate School of Information Science and Technology, the University of Tokyo.

3.2 Design

The participants conducted the main experiment in four conditions. In each condition, the participants were instructed to practice a provided speech script written on paper. While the participants practiced their speech, at chosen times, the participants were instructed to record their speech. The participants could listen to the most recent recording of their speech as many times as they wanted while practicing their speech under the specified condition. In each condition, the participants could take notes on the script and practice aloud even when they were not recording themselves.

We prepared four different speech scripts in which one script was used for one condition. The participants were informed that each script was a segment of a student commencement speech given at the participants' alma mater's commencement ceremony.

3.2.1 Conditions. The main experiment followed a 2 x 2 within-subject factorial design. The two independent variables were voice manipulation (with, without) and model audio (with, without). The participants completed four conditions: 1) Condition M/A (with voice manipulation, with model audio), 2) Condition M/nA (with voice manipulation, without model audio), 3) Condition nM/A (without voice manipulation, with model audio), 4) Condition nM/nA (without voice manipulation, without model audio). The two independent variables are detailed below.

Voice Manipulation. In the conditions with voice manipulation (Condition M/A and Condition M/nA), the participants manipulated their recorded audio using Praat, a computer software used for speech analysis in phonetics, which allowed the participants to record, save, and listen to audio, and manipulate the vocal parameters of the recorded audio.

Once the participants recorded their speech using Praat and inputted it into Praat's manipulation view, Praat visualized the audio's sound pulses, pitch values from the sound pulses, and the duration of sound, as shown in Figure 1. The sound pulses, which could not be manipulated, indicated where there was sound in the audio (Figure 1 top area). The participants could manipulate the pitch and duration of the voice input.

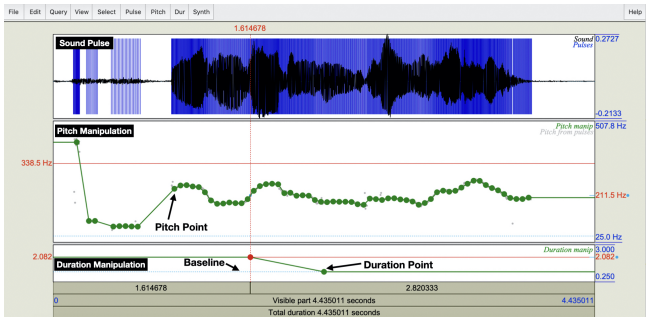


Figure 1: Example screen view of voice manipulation using Praat.

The audio's pitch was visualized with pitch points (Figure 1 center area), where each pitch point corresponded to the pitch value (Hertz) at a specific moment in the audio. The participants could manipulate the pitch value of one pitch point or multiple pitch points that were between two moments in the audio, by manually moving the pitch point vertically or by entering a pitch value that indicated the change in pitch.

The participants could also manipulate the duration of the audio (Figure 1 bottom area), by adding and moving duration points which indicated the relative duration of the audio. The participants moved a duration point horizontally to specify the moment in the audio that they wanted to change the duration of, and moved the point vertically to alter the duration. The original audio's duration point value was set to a baseline of the value 1.00; the duration points that were moved above the baseline (duration point value>1.00) lengthened the duration, and the duration points that were moved below the baseline duration point value<1.00) shortened the duration.

Model Audio. The model audio was a recorded segment of a student commencement speaker giving a commencement speech at the participants' alma mater. The speakers of the model audio were members of a predominantly female community and had vocal qualities that were presumed to be those of females. We transcribed the recorded segments and provided them as speech scripts to the participants. Two model audio were used for the main experiment

because there were two conditions with model audio (Condition M/A and Condition nM/A).

3.2.2 Measurements. All measurements described below were gathered through questionnaires.

Initial Session Questionnaire. In an initial session questionnaire, the participants evaluated their self-esteem and self-efficacy, which have been closely studied with task performance and achievement [14, 17, 23]. To measure self-esteem, the participants used the Rosenberg Self-Esteem Scale [38], and to measure self-efficacy, the participants used The GSE Scale [43].

The participants also provided self-assessments across four items that reflected their perceptions of their voice and speech performance: *confidence, satisfaction, communication, self-expression*. We collectively refer to these four items as "four areas of self-perception." First, the item of *confidence* was informed by prior work that has shown the positive impact of role models of public speaking on self-efficacy and confidence in public speaking [42]. However, there has not been much work on the impact of user-controlled voice manipulation on confidence. Second, *satisfaction* was selected because the relation between confidence and satisfaction in public speaking has not been studied thoroughly. Third, the item of *communication* was measured to help us understand the influence of listening to role models of public speaking and voice manipulation on the perceived ability to communicate, which has not been extensively studied in prior work. Fourth, *self-expression* was chosen because we are interested in investigating whether the speech practice techniques in this experiment will influence the participants' perceived abilities to express themselves through their voice and speech.

The participants rated one statement for each of the four areas of self-perception on a 7-point Likert scale on a continuum from 1: strongly disagree to 7: strongly agree (Table 1). To distinguish *confidence* from *satisfaction*, participants were told that "*confidence* is the ability to effectively perform a task whereas *satisfaction* is the fulfillment of happiness (i.e. 'are you happy with what you are evaluating?')." The same questionnaire items were used in the main experiment and final session questionnaire.

Table 1: Questionnaire items used in the initial session, main experiment, and final session to measure self-perceptions of confidence, satisfaction, communication, and self-expression.

Area of Self-Perception	Statement
Confidence	I am confident in my voice and the way I speak.
Satisfaction	I am satisfied with my voice and the way I speak.
Communication	I can communicate what I want to say or convey to the listener, through my voice and the way I speak.
Self-Expression	I can express myself through my voice and the way I speak.

Main Experiment. In the main experiment, before and after each condition was conducted, the participants rated the statements for each of the four areas of self-perception (Table 1). There were nine scales that the participants rated for each of *confidence* and *satisfaction*: overall voice, loudness, pitch, speed, intonation, articulation, tone, quality, flow. These nine scales were informed by vocal qualities referred to in previous research on voice [40, 44]. The participants rated one statement each for *communication* and *self-expression*.

Final Session Questionnaire. At the end of the main experiment, the participants answered a final session questionnaire, which consisted of three items. The first item asked the participants to rate how ideal the model audios' speech deliveries were, and to elaborate on their ratings in written responses. We refer to this evaluation as "the quality of model audio." Ratings were conducted on a 7-point Likert scale on a continuum from 1: strongly disagree to 7: strongly agree (Table 2 top). Next, the participants were asked to select which of the four conditions contributed the most to each of the four areas of self-perception (Table 2 middle). The participant was asked to elaborate on their choices in written responses. Finally, the questionnaire asked the participants to evaluate statements regarding the four areas of self-perception (Table 2 bottom).

Table 2: Questionnaire items used in the final session questionnaire.

Item	Statement
Quality of Model Audio	1) The first model audio is a speech that I consider to be the ideal speech. 2) The second model audio is a speech that I consider to be the ideal speech. 3) (written) Please elaborate on your responses.
Comparison of Conditions	1) Among the 4 conditions, which made you feel most confident in your voice and the way you speak? 2) Among the 4 conditions, which made you feel most satisfied with your voice and the way you speak? 3) Among the 4 experiments, which were you able to communicate what you want to say or convey, with your voice and the way you speak? 4) Among the 4 conditions, which were you able to express yourself the most, with your voice and the way you speak? 5) (written) Please elaborate on your responses.
Four Areas of Self-Perception	Refer to Table 1.

3.2.3 Semi-Structured Interview. To investigate **RQ3**, we performed semi-structured interviews after the participants completed the final session questionnaire. The interview session was composed of two parts: an *experiment-related* inquiry asking about the participants' experiences with voice manipulation and the model audio, and a *social* inquiry asking about personal experiences that affected the participants' self-perceptions of their voice and speech. The aim of the *experiment-related* was to understand how voice manipulation and the model audio affected the participants' self-perceptions. We also inquired about how the participants used voice manipulation and model audio to practice their speeches. The questions asked during the *social* inquiry were tailored to the participants' previous responses in the interview, and their willingness to share information. Interviews were audio-recorded and transcribed. The average time of the 18 interviews was 12 minutes and 43 seconds ± 2 minutes and 52 seconds (SD).

3.3 Hypotheses

We draw from previous work to establish our hypotheses which were developed to answer **RQ1** and **RQ2**.

Research has shown that the confidence in the ability to speak impacts the perceived ability to speak [8], but that public speaking exercises increase satisfaction but decrease confidence [31]. Based on this work, we make the following hypothesis:

[H1] Listening to the model audio and using voice manipulation, individually and combined, during speech practice, increase the satisfaction in voice and speech but decrease confidence.

However, we note that role models are effective when they have similar attributes to the learner [6, 24, 25], and so the relatability of the model audio may impact the participants' confidence and satisfaction in their voice and speech.

By listening to the model audio, participants will understand what a communicative speech sounds like, and will attempt to emulate the communicative aspects of the model audio. Voice manipulation alone will not help the participants determine if they are communicating effectively. We make the following hypothesis:

[H2] Listening to the model audio alone or using the model audio with voice manipulation increases the perceived ability to communicate.

By comparing their voices to those of the model audio, the participants will recognize how their voices are unique. Through voice manipulation, the participants will explore their unique vocal parameters. Based on this, we make the following hypothesis:

[H3] Listening to the model audio and using voice manipulation, individually and combined, during speech practice, increase the perceived ability to self-express through voice and speech.

3.4 Procedure

The experiment had three stages - initial session, main experiment and final session - as described in Figure 2.

The experiment was divided into two consecutive days, in which the average time for each day was approximately one hour. On day one, the participants completed an initial session questionnaire and conducted two of the four conditions. On day two, the participants conducted the remaining two conditions, completed a final session questionnaire, and engaged in semi-structured interviews.

At the beginning of day one, the participants completed an initial session questionnaire, in which the participants self-assessed their self-esteem and self-efficacy and the four areas of self-perception (Table 1). This was followed by a two minute warm-up session, in which the participants used a practice script to practice reading and speaking it aloud. This was followed by an introduction to Praat and instructions on manipulating pitch and duration. The participants were given time to practice using Praat to manipulate the pitch and duration of a sample audio.

The participants then began the main experiment. The procedure of conducting one condition in the main experiment is described in Figure 2 (center). Each condition began with one minute to practice reading aloud and taking notes on a provided speech script to prepare for an initial recording of their speech that followed. The same script was used for the entirety of conducting one condition. In the conditions with model audio (Condition nM/A and Condition M/A), the participants listened to the model audio twice before the one minute practice. After one minute passed, the participants created an initial recording of their speech, listened to it, and then completed an initial recording questionnaire (Table 1). The participants were then instructed to practice their speech for 10 minutes. The participants were instructed to aim to speak in a way that made them most confident and satisfied in their voice and speech, and so that their speech was communicative and expressed themselves. During the 10 minutes, the participants conducted three rounds of speech practice. Each round was 200 seconds. At the end of each round, the participants re-recorded their speeches. A set time for re-recording was established to control the number of re-recordings. The purpose of re-recording was to allow the participants to read aloud the entire script and hear how their speeches sounded after each round. During the second and third rounds, the participants listened to the re-recording from the end of the previous round as many times as they wanted, and continued to practice their speech. During the first round, the participants listened to the initial recording. In the conditions with voice manipulation (Condition M/nA, Condition M/A), during each round, the participants manipulated the most recent recording - either the initial recording (during the first round) or the re-recording from the end of the previous round (during the second and third rounds) - and visually observed and heard the results of manipulation (i.e. AAF) using Praat. At the end of the third round, the participants created a final recording of their speeches. Upon listening to the final recording, the participants completed a final recording questionnaire with the same instructions and items as the initial recording questionnaire (Table 1).

The order in which the participant executed the four conditions were randomized. The order of the four scripts was the same for all participants. After completing all four conditions, the participants answered the final session questionnaire (Table 2), which was followed by a semi-structured interview.

4 RESULTS

4.1 Self-Esteem and Self-Efficacy

We calculated the participants' self-assessments of self-esteem using the Rosenberg Self-Esteem Scale [38], where each participant's total score can range from 10 to 40. The results revealed that the median score was 28 and the mean score was 28.22 ± 4.87 (SD). We also calculated the participants' ratings of self-efficacy using The GSE Scale [43], where each participant's total score can range from 10 to 40. The results showed that the median score was 30 and the mean was 30.44 ± 4.31 (SD).

4.2 Main Experiment

For each of the four conditions conducted in the main experiment, we calculated the differences between each participant's ratings of *confidence, satisfaction, communication, self-expression* from the initial recording questionnaire and the final recording questionnaire. We refer to these calculated values as "score differences." We calculated the score differences for all scales in each of the areas of self-perception (*confidence*: nine scales, *satisfaction*: nine scales, *communication*: one scale, *self-expression*: one scale).

For each scale, because Likert scale is an ordinal scale, we applied aligned-rank transform (ART) using ARTool, which allowed the use

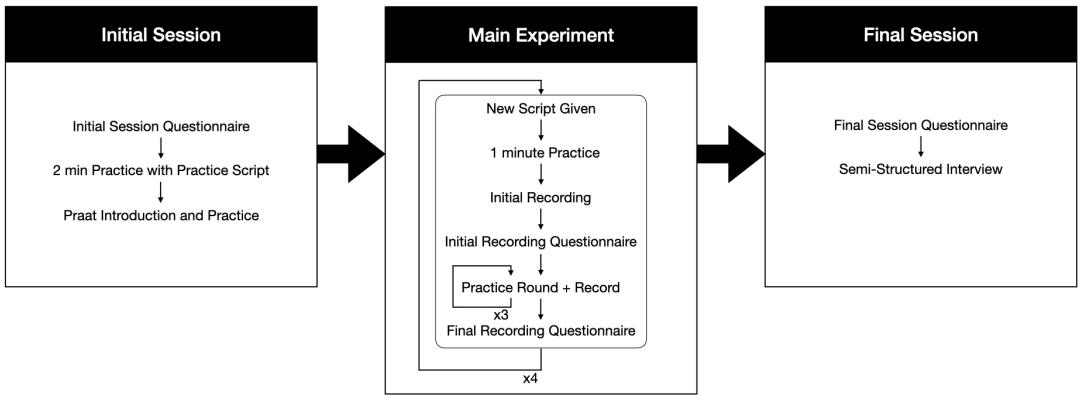


Figure 2: Diagram of experimental flow.

of ANOVA to analyze the interaction effects with non-parametric data [56], and conducted two-way ANOVA. Our analysis considered the within-group factors of model audio (two levels: with, without) and voice manipulation (two levels: with, without). When there were significant interaction effects, we conducted post-hoc pairwise comparisons using the Wilcoxon signed-rank test, applying the Holm correction considering the within-group factors of model audio and voice manipulation.

4.2.1 Confidence. The results of all nine scales of *confidence* are shown in Figure 3. There was a significant main effect of the model audio on tone ($F(1, 17) = 4.74, p < .05$) (Figure 3 bottom left); the score difference of tone in conditions with model audio (Median=1) was significantly higher than that the score difference in conditions without model audio (Median=0). There was no significant main effect of the model audio on the other eight scales. There was no significant main effect of voice manipulation on all nine scales. We observed a marginal significant interaction effect on flow ($F(1, 17) = 3.92, p = .06$). Nevertheless, the results of post hoc tests for pairwise comparisons revealed that there were no significant differences. We did not observe significant interaction effects on the other eight scales.

4.2.2 Satisfaction. The results of all nine scales of *satisfaction* are shown in Figure 4. The results of ANOVA revealed that there were no significant main effects on the model audio and voice manipulation on all nine scales. We found that the interaction effect was significant for the scale of overall voice ($F(1, 17) = 4.96, p < .05$) and that there was a marginally significant interaction effect of quality ($F(1, 17) = 4.32, p = .05$). Nevertheless, the results of post hoc tests showed that there were no significant differences for either of the scales. The other seven scales did not have significant interaction effects. Taken together with the results of *confidence*, **[H1]** was not supported.

4.2.3 Communication. We did not find any significant effects for *communication* (Figure 5 left), which did not support **[H2]**.

4.2.4 Self-Expression. There were no significant main effects on voice manipulation and model audio. The results revealed that there was a significant interaction effect ($F(1, 17) = 4.96, p < .01$) (Figure 5 right). The results of a post hoc test showed that there was a trend for the score difference in Condition M/nA to be higher than that of Condition nM/nA ($p = .09$), but the results were not strong enough to support **[H3]**.

4.3 Comparison of Conditions

Based on the responses from the final session questionnaire, in all four areas of self-perception, the most number of participants selected Condition M/A (with voice manipulation, with model audio) as the condition that was most effective for increasing their self-perceptions in each area of self-perception. The distribution of responses is shown in Figure 6.

4.4 Quality of Model Audio

The results of the ratings of the model audio from the final session questionnaire revealed that the median score was 6. This means that the participants considered the speech delivery and voices of the speakers of the two model audio close to ideal. The quality of model audio is referred to again in the results of the thematic analysis.

4.5 Change in Self-Perceptions from Initial Session to Final Session

We conducted pairwise comparisons between the scores of *confidence, satisfaction, communication, self-expression* from the initial session questionnaire and the scores from the final session questionnaire. We used the Wilcoxon signed-rank test, applying the Holm correction. The results showed that the score difference was significant only for *satisfaction* ($Z = -2.67, p < .05$); the scores in the final session questionnaire (Median=6) were significantly higher than those in the initial session questionnaire (Median=4). The score differences between the ratings of *confidence, satisfaction, communication, self-expression* between the initial session questionnaire and the final session questionnaire are shown in Fig 7.

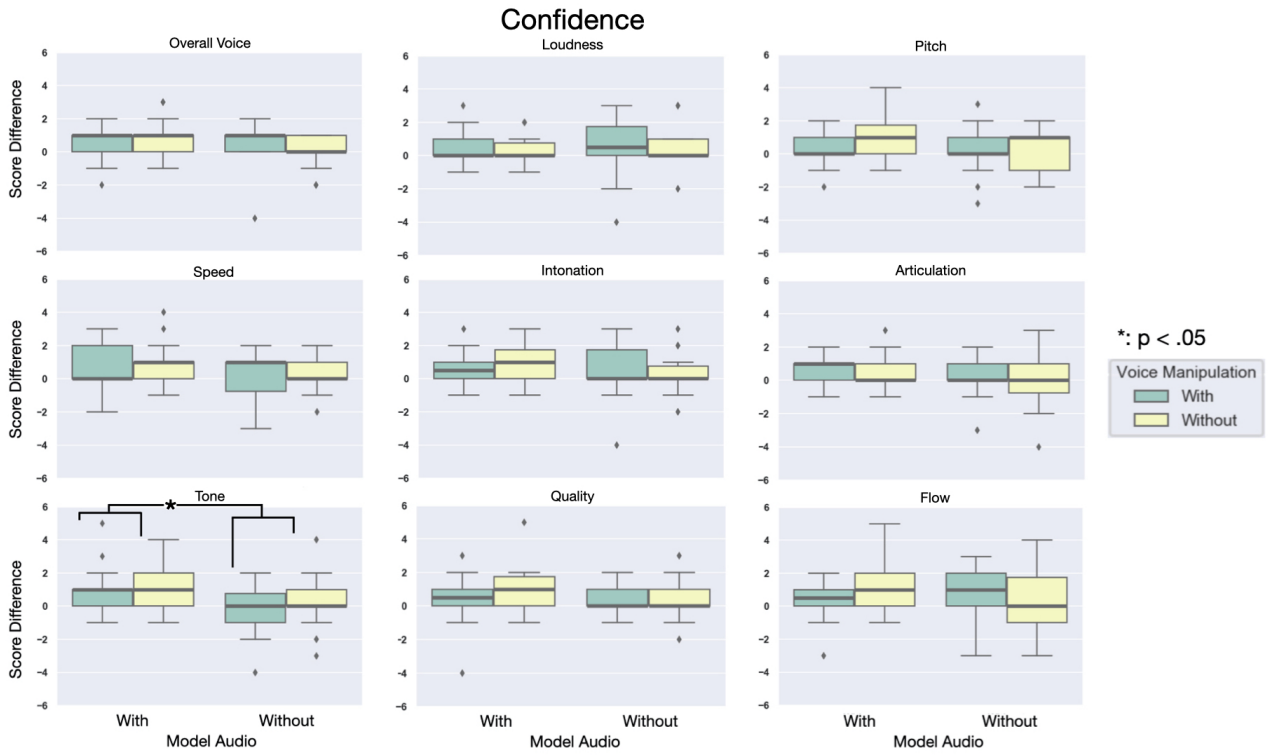


Figure 3: Box plots of the score differences in the nine scales of confidence in the four conditions.

4.6 Themes

We conducted a thematic analysis of the data collected from the semi-structured interviews and written responses in the final session questionnaire using a mixture of inductive and deductive coding, as outlined by Braun and Clarke [9]. After creating initial codes, we searched for themes from the codes, reviewed themes, and defined and named themes. If the same data was applicable to more than one code, it was tagged with all appropriate codes. The codes were eventually grouped into five high-level themes: Effectiveness of Model Audio, Effectiveness of Voice Manipulation, Note-Taking for Recollection, Acceptance of Own Voice, and The Influence of Gender on Voice.

4.6.1 Effectiveness of Model Audio. This theme is composed of data that explain how the participants evaluated the effectiveness of the model audio for improving speech performances. 17 participants expressed that the model audio was helpful for practicing their speeches.

Comparing Own Voice to Model Audio. 10 participants said that they compared their voice to those of the model audio. Among these participants, five participants created comparisons to determine what aspects of the model audio they wanted to adopt into their own voices. P4 said that they explored their voice more by comparing their speech delivery to the model audio's, whereas practicing a speech without the model audio was more difficult: "*Having the models was more fun for me because I could then learn about myself by comparison, versus when I'm just by myself, it's a little harder to catch what my habits are or what the other possibilities are…*"

Six participants said that they felt that the model audio negatively influenced their self-perceptions of their voices, because the model audio had better speech deliveries than themselves. Five of these participants said that the model audio was nevertheless helpful for speech practice, as summarized by P1: "*I think the voice model made me less confident initially in the way I speak, but when I used it to practice, I felt like I would be more confident if I was to give the speech, since it helped me iterate on my own flaws in speech.*" Only one participant, P15, preferred to practice their speeches without the model audio: "*...[the condition of] no model was easier for practicing because then I don't have to compare myself to [the model audio]...with that model audio it's added pressure of almost trying to imitate it.*"

Quality of Model Audio. The participants said that the effectiveness of the model audio was contingent on the model audio's speech performances. 14 participants attended to the vocal qualities presented in the model audio - nine mentioned tone, eight referred to speed, six discussed intonation, four mentioned pitch, four considered flow, and two referred to the articulation of the model audio when discussing their evaluations of the model audio. P11 described the different vocal qualities that they noticed between the two model audio that they listened to: "*The first model audio was confident and conveyed meaning and expression very well through both intonation, speed, and fluctuation in their style of speaking. The second one seemed to have more of an even and consistent tone, which was good, but varied/ fluctuated less and therefore seemed to convey*

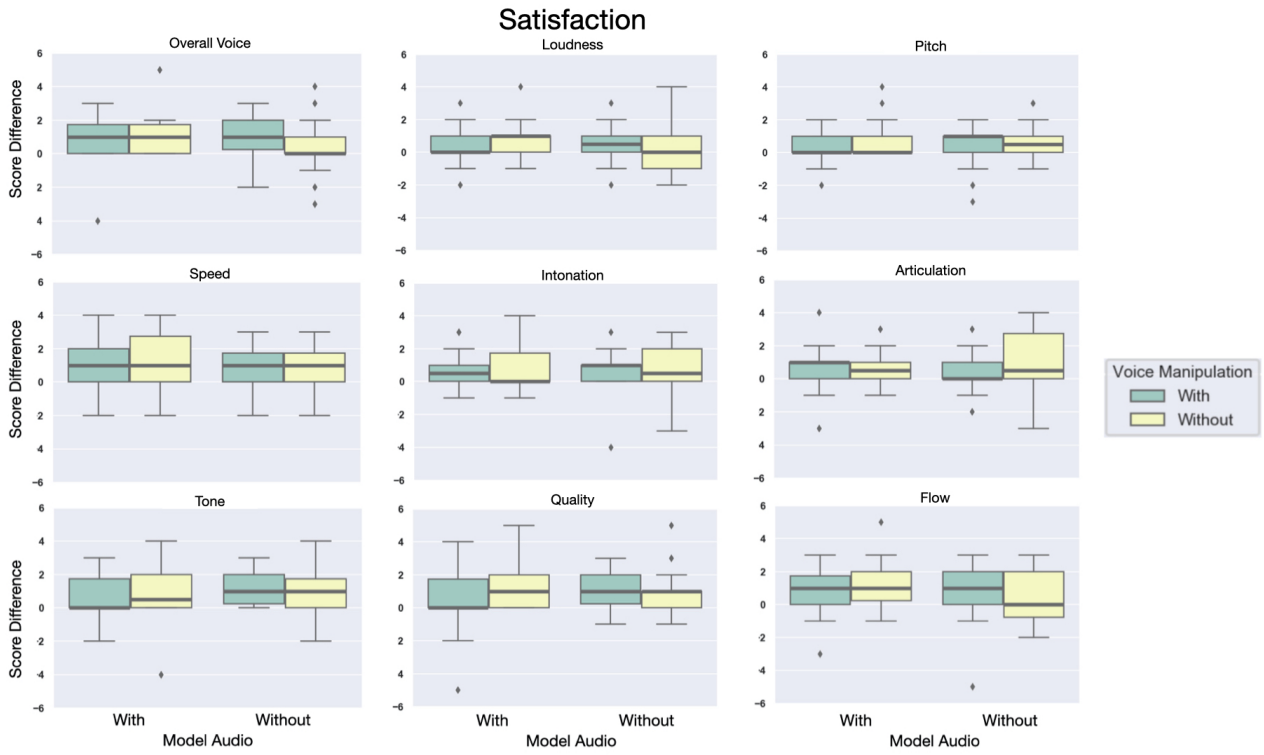


Figure 4: Box plots of the score differences in the nine scales of satisfaction in the four conditions.

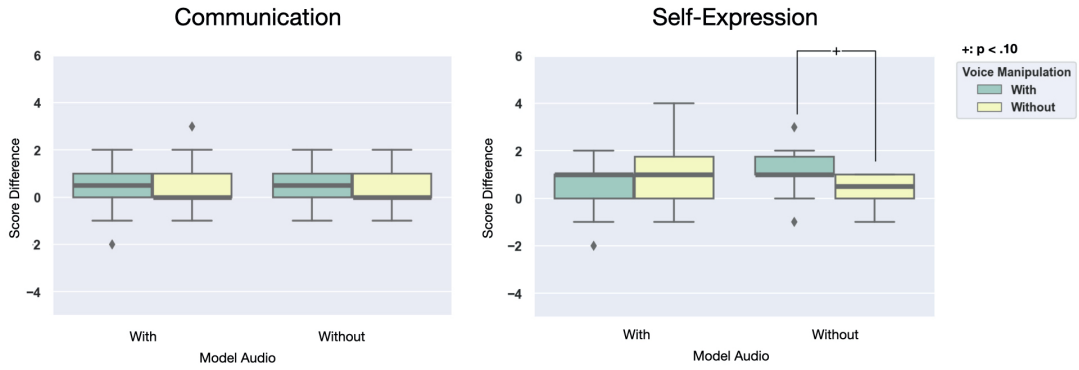


Figure 5: Box plots of the score differences in communication (left) and self-expression (right) in the four conditions.

less expression and be less engaging." Four participants said that their ability to relate to the speaker of the model audio influenced whether or not the participants considered the model audio as ideal speakers, as summarized by P4: "*I appreciated the feminine, high-pitch, and the joyfulness of a graduating student shining through, but did not identify with that character.*" Four participants indicated that authenticity and genuineness in the model audio's speech delivery was critical for public speaking. P1 implied that the speaker's ability to connect with their words enhanced the authenticity of the speech: "*[The model audio] yesterday sounded less heartfelt, more practiced. The one today felt more heartfelt, felt more sincere and connected with the words more.*"

4.6.2 Effectiveness of Voice Manipulation. This theme describes how the participants approached voice manipulation, how voice manipulation helped with or failed to help with speech practice, and suggestions made by the participants for improving the voice manipulation experience. 11 participants expressed that voice manipulation was helpful for speech practice. Four participants said that they experienced initial hesitation with using Praat or that they wished they had more time to familiarize themselves with the software.

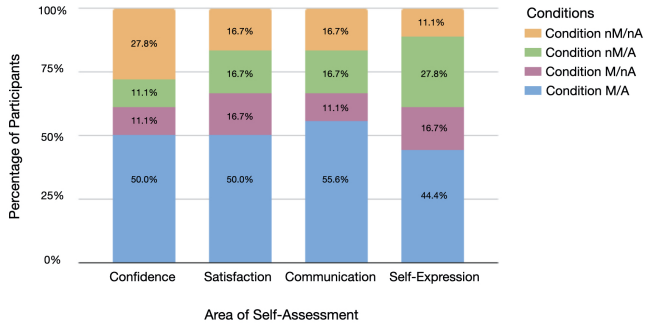


Figure 6: Distribution of the four conditions that contributed the most to self-perceptions of confidence, satisfaction, communication, and self-expression.

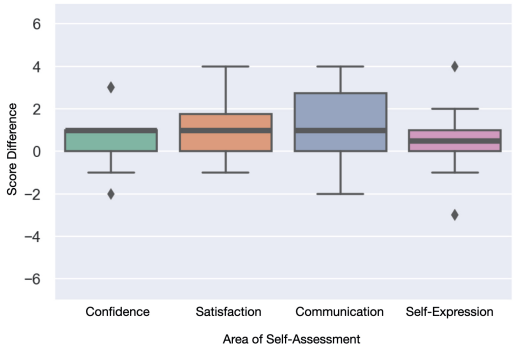


Figure 7: Box plots of the score differences in the ratings of confidence, satisfaction, communication, and self-expression between the initial session questionnaire and the final session questionnaire.

Preview of How Voice will Sound. Eight participants said that they interpreted the voice manipulation as a means to gain a preview of how their voices would sound with vocal modulation. Among these participants, four said that they gained different perspectives of their voices, because listening to the AAF of their manipulated voices was similar to listening to a different speaker's voice, as explained by P18: "*I liked having the manipulation because it's similar to having the model [audio] there, but I can hear it in my own voice. And so that way, I had an example in my own voice, so my brain is like, 'oh I can do this. I've already heard myself doing [it].*'" Two participants positively evaluated the immediate feedback they gained from voice manipulation, as summarized by P6: "*[When] I'm practicing it out loud...I don't get that immediate feedback, unless it's with the manipulation, so I'd say [practicing my speech out loud] was helpful but probably not as helpful as [receiving] immediate feedback.*"

Identifying Changes that Need to be Made. Seven participants said that voice manipulation helped them identify the changes they needed to make to improve their voices and speech delivery. P2 said that voice manipulation helped them locate parts in the script that they wanted to sound different in: "*[Voice manipulation] was helpful because then [I could] pinpoint what would make [my voice and speech] sound better.*" P14 implied that with voice manipulation, they explored various possibilities of how their voice could sound: "*I was able to adjust and hear how the speech would sound slower and [in] a deeper or higher voice. I was able to take notes from [what I heard in voice manipulation] and apply it to my final version...*"

Limitations of Voice Manipulation. Six participants explicitly said that voice manipulation was not helpful. Within these participants, four participants explained that they were unable to fully explore the possibilities of vocal modulation because Praat only allowed pitch and duration manipulation, as summarized by P10: "*I didn't really find the software that helpful because I was focusing on being more expressive and articulate which aren't things that the software can alter.*" Four participants described voice manipulation as a helpful technique with the caveat that they could not rely on voice manipulation alone to improve their voices and speeches. These participants explained that while voice manipulation was helpful, they could have also practiced their speeches without it, as summarized by P14: "*I feel like I wouldn't depend on it by itself but I feel like it was a nice trait to use to like see where I could make adjustments and changes.*"

Receiving Immediate Feedback through Visualization. Four participants said that the visualizations of their vocal parameters were helpful for learning about their speaking tendencies and observing how voice manipulation changed their vocal parameters. P14 said they were able to examine the pitch values of their voice, which would be difficult without Praat: "*You could physically see where your voice is higher and like taking note of that like 'oh do I want that one point to be a bit higher a bit lower?'...[Identifying parts to change the pitch] would be difficult if [the voice manipulation software] didn't have [the visualization of my pitch].*"

4.6.3 Note-Taking for Recollection. This theme describes how participants interpreted the purpose of note-taking during speech practice. Nine participants mentioned note-taking as an effective technique for recalling the changes they wanted to make during speech practice. Within these participants, six mentioned that note-taking was helpful when done with listening to the model audio. P18 explained that note-taking was effective for reminding themselves to not imitate the model audio when they did not like the way the model audio sounded: "*I would make a note of it on the paper so that way I would remember like 'don't emulate the model here.*'" Two participants said they took notes while using voice manipulation to remind themselves of the manipulation techniques they used in Praat, as summarized by P11: "*...manipulation was trying to figure out what I wanted. And then the note-taking was more just a recollection of how [I] would make that happen.*"

4.6.4 Acceptance of Own Voice. This theme reflects the sentiments of eight participants who gradually came to terms with their voices through aspects of the experiment. P18 said that their difficulty with identifying their accent had caused internal confusion, but the opportunity to repeatedly listen to their voice was effective for clarifying their confusion: "*[The experiment] kind of helped me realize that I still have control over the way I talk. I was also having some kind of identity crisis of 'what kind of accent do I have?' I grew up in Appalachia, but I came here from the South...[the experiment]*

helped me realize [that I have a] pretty standard American accent." P18 implied that their accent is tethered to their identity and background, and listening to their accent objectively helped them clarify their style of speaking. While not included in this theme, quotes from four other participants mentioned accents, saying that these participants avoided talking in particular styles, namely Southern accents and the Valley Girl dialect, because they recognized that there are negative connotations of these speaking styles. The participants implied that society can influence the way in which they perceive accents.

Five participants mentioned that while they had concerns about their voices prior to the experiment, they gained more confidence and satisfaction in their voices after listening to their voices and practicing their speeches. Among these participants, P2 said that the experiment helped them recognize areas that required improvement, but acknowledged that the previous concerns they had with their voice were alleviated: "*Before, I thought my slightly lower voice was kind of irritating...but now I'm like, 'honestly, it sounds fine,' so I have no problem with it...I think that [the experiment] kind of pointed [my slow speaking] out a little bit...I have improved in some way but I definitely have a tendency to speak kind of quickly.*"

4.6.5 The Influence of Gender on Voice. This theme includes comments from the participants that discussed how gender influences the way they perceive their voices and public speaking. Six participants discussed the relationship between gender and voice, and in particular gender-related stereotypes and the pressure to modulate their voices.

Gender-Related Stereotypes and Voice. Three participants elaborated on vocal stereotypes related to gender. P2 explained that men's voices are generally found to be more convincing and authoritative, while women's voices are not: "*...the way that men speak generally is considered to be fairly, um, convincing or authoritative in a way...Maybe in society, we simply view women to speak less authoritatively maybe with higher pitch voices... with more floral tones and just qualities that where maybe it's just more flowery or soft, and like basically just [in] less authoritative and commanding ways.*" P5 explained that while they have a high-pitched voice, they experience discontinuity because they do not relate to the characteristics associated with higher-pitched voices, which are most commonly tied to femininity: "*High voices are associated with youth and femininity which are things that in general I am not good at performing. So I always am like 'does my voice seem discontinuous with the rest of me.?'...High voices are associated with just being naive and stupid...of course that's not true but it's really hard to kind of deprogram that.*"

Social Pressure to Emulate a Man's Voice. Four participants said they felt the pressure to emulate men's voices while speaking in professional settings. P5 implied that the pressure to sound like a man stems from society's tendency to associate the ideal form of public speaking with men: "*All of the people I think of as really good speakers like you know, Martin Luther King like presidents...those are men. The people that I'm sort of taught to hold up as like paragons of the speech, that's men.*" P14 explained that the consequence of not emulating men's deeper voices is exclusion from the professional sphere: "*In general, women are being trained like 'oh you have to speak deeper to be taken more seriously and if you have a high pitched voice...maybe the professional world isn't like the right place for you.'*"

Coming to Terms with Own Voice Without Following Stereotypes. Two participants explained that while they and other women are conditioned to think that speaking like men is a requirement to be taken seriously, they recognized that people can give convincing speeches without being restricted by vocal stereotypes, as summarized by P2: "*You can still give a very convincing speech and like not sound like a man basically, and like I feel like it'd be interesting to see in what ways like the female voice can...I guess like shift or in what ways it can sound more convincing and authoritative without having to sound like a man.*"

5 DISCUSSION

The main findings are as follows:

(1) The results of ANOVA did not support any of our hypotheses. However, the results showed that listening to the model audio while practicing a speech increased the confidence in tone ($p < .05$). There was a trend for the self-perception of self-expression to increase after using voice manipulation without the model audio, than without using voice manipulation without the model audio ($p = .09$).
(2) The results of our thematic analysis showed that 11 participants and 17 participants highly evaluated the usage of voice manipulation and the model audio, respectively. The most number of participants selected Condition M/A as the condition that contributed the most to improving all four areas of self-perception. We infer that there may be other factors besides using or not using voice manipulation and/or the model audio that influenced the participants' self-perceptions.
(3) The participants' backgrounds and social experiences seemed to impact the way the participants perceived their voices. Discussion of gender-related stereotypes and self-perception of accents were prevalent, indicating that there were participants who reflected on how social factors influenced the way they speak.

We detail our findings with respect to the three hypotheses.

[H1] The results showed that there was a significant positive impact of listening to the model audio on the confidence in tone. The most number of participants (9/18) considered the tone of the model audio when determining whether or not to take inspiration from the model audio, so it is possible that the participants referred to positive examples of tone in the model audio and applied them to their speech practice. However, listening to the model audio and/or using voice manipulation did not have significant effects on confidence and satisfaction in all vocal qualities, which did not support **[H1]**. We infer three possible explanations. First, the participants' confidence and satisfaction fluctuated during the usage of the model audio, because they experienced a decrease in confidence and satisfaction when they compared themselves to the model audio, even if the model audio eventually assisted them with speech practice. Second, with respect to voice manipulation, the participants could not improve certain vocal qualities (e.g. expressiveness, articulation) because they were unsure of how to improve those with the manipulation of pitch and duration. Because these participants could not improve the vocal qualities they wanted to change the most,

their confidence and satisfaction in their voice and speech after using voice manipulation was not significant. Third, listening to the model audio and voice manipulation did not significantly affect the participants' confidence and satisfaction because the participants had high initial levels of self-esteem and self-efficacy, which are associated with satisfaction [18, 19, 27] and confidence, [28, 32, 49] respectively. As seen in Table 7, the participants' confidence and satisfaction did not drastically change from the beginning of the experiment to the end.

[H2] The impact of the model audio on the perceived ability to communicate was also inconclusive, which was contrary to **[H2]**. Although the participants considered the model audio's vocal qualities while practicing their speeches to improve communication, it is possible that the participants' ability to resonate with the speech itself had a greater impact on their perceived ability to communicate. Since the participants did not write their own speeches in this study, it is likely that there were instances in which the participants could not resonate with the given speech script.

[H3] We could not observe any significant effects of listening to the model audio and/or using voice manipulation on the perceived ability to self-express, contrary to **[H3]**. However, the results showed a trend for the score difference in self-expression in Condition M/nA to be higher than that of Condition nM/nA, which implied that using voice manipulation without listening to the model audio may have improved the self-perception of self-expression. One possible explanation for this trend may be that during voice manipulation, the participants chose the vocal parameters to manipulate and maintained the original vocal parameters that they preferred not to change, which in turn preserved their self-expression. Another explanation is that listening to the AAF of their manipulated voices resembled the experience of listening to the model audio, as described by participants. Participants explained that the degree to which they relied on the model audio was influenced by whether or not the model audio presented similar attributes as the participants, which align with the findings of prior work [6, 24, 25]. It is possible that the participants found familiarity with the AAF of their manipulated voices, which in turn may have encouraged effective vicarious experiences [6] that was effective for enhancing self-expression.

Lastly, while not all participants mentioned gender in the context of public speaking and voice, it is important to discuss the social implications of our findings pertaining to gender. There were participants who recognized that they had been conditioned to emulate men's deeper, more authoritative voices, which reflect the findings of previous work [46]. We also found that the participants felt that the inability to speak like men resulted in exclusion from the professional world, which is an experience that has been recognized in prior work [7]. These findings suggest that external assessments of public speaking have the power to reinforce a style of speaking that favors men's voices. Deprogramming the association of the ideal form of public speaking with men's voices is difficult. However, as the participants explained, it is important to consider how people with higher-pitched voices can train their natural voices to be considered appropriate for public speaking, without having to make unnatural vocal modulations. One approach may be to shift away from speech training techniques that condition people with higher-pitched voices to sound more like men (e.g. using speakers who are men as examples of ideal public speakers). Given our findings that the model audio was most effective for speech practice when they were relatable, it is worth considering how learning from speakers who have similar identities and attributes as the learner can positively influence the learner's self-perception and speech performance.

5.1 Design Implications for Speech Practice Systems

Overall, our results imply that voice manipulation and model audio has the potential to increase self-perceptions of voice and speech. However, this can be better validated if certain considerations are made in the design of a speech practice system that incorporates voice manipulation and model audio. We draw on our findings and existing speech practice systems to propose design considerations for a speech practice system to support users in improving their self-perceptions of voice and speech.

5.1.1 User-Controlled Voice Manipulation. The participants implied that they manipulated vocal parameters of their choosing, which we infer is due to Praat's flexibility of voice manipulation. While there was a participant who preferred not to manipulate their pitch because their pitch was central to the uniqueness of their voice, another participant said pitch manipulation allowed them to incorporate more self-expression into their speech. Our findings are similar to those of previous work that found that users preferred to hear the resynthesis of certain vocal parameters over others [39]. Based on our findings and existing work, the resynthesis of voice could be more effective if users could select the vocal parameters that they wish to change. As discussed earlier, hearing the AAF of their manipulated voices helped convince the participants that they could improve their voices and speeches. It is important to give users the agency to choose how to change their voice and allow them the chance to self-assess whether those changes are appropriate for the users' purposes. This is corroborated by our findings that the participants felt that they had control over the way they talk and were able to alleviate frustrations they had with respect to their voices. Supporting users in the process of resynthesizing their voices might also help users gain knowledge on voice modulation, which has a great influence on engaging listeners and delivering presentations [26]. However, some participants indicated that they wished they had known how to improve articulation or expressiveness, and struggled to improve these vocal qualities because Praat did not provide guidance. It may be helpful for a speech practice system to offer knowledge on the effect of vocal modulation of certain vocal qualities.

5.1.2 Model Audio with Similar Attributes as the User. As indicated by some participants, the model audio pressured the participants to imitate the speech delivery of the model audio, which implied that the model audio had a persuasive influence, which was negative for some participants. Still, all but one of the participants expressed that the model audio was effective for speech practice. Taken together, we infer that the participants preferred to receive guidance from the model audio on how to improve their speech, to the degree that it did not impose a way of speaking.

A speech practice system should recommend model audio that presents similar characteristics to the user and traits that the user wants to acquire. The participants in our study attended to the similarities between their voice and those of the model audio, and attempted to emulate the model audio when it had favorable vocal traits. One way a speech practice system could recommend a desirable model audio is by taking an initial measurement of the user's vocal parameters (e.g. pitch, volume), and matching the user with an appropriate model audio that shares similar vocal qualities. It may also be helpful for the user to specify the context of their speech, so that the system could better match the user to a model audio of a speaker in a similar setting with a similar purpose, which is a feature observed in previous work [55]. The participants also expressed that they would be interested in learning public speaking skills from women speakers, because they have not had much public speaking education based on women role models of speaking. These sentiments are also reflected in prior work that found that women speakers who were exposed to women role models made improvements in their oral presentations and had more positive self-evaluations of their speeches [24]. We suggest that a speech practice system offer a collection of model audio that is diverse and use non-cissexist language when referring to voice types [2]. This could expand the opportunities for users to find a role model of public speaking that they can relate to.

5.1.3 Annotations. The participants expressed that taking notes on the paper copy of the script was effective for recalling how the model audio spoke and reenacting the AAF, implying that their notes reminded them of the changes they needed to practice. The flexibility to incorporate individual note-taking methods supported the participants in preparing to speak their speeches out loud. Research has also shown that creating speaking outline notes is among the most important activities in speaking preparation that predicts high evaluations from listeners [34]. Including a feature that allows users to take notes on the script produced by the speech-to-text service could especially be meaningful for users who experience communication apprehension [12, 34].

5.2 Limitations

The participants could not modify the provided speech scripts and only conducted public speaking training by themselves and not in front of potential listeners. We cannot easily extend the results of this study to situations in which the participants write their own speeches and speak in front of live audiences. Additionally, despite the randomization of the order of the four conditions in the main experiment, there may have been ordering effects on the results.

In this study, we did not recruit men; all but one participant identified as a cisgender woman. The small sample size and gender distribution of the participants may have limited the scope of our findings on the topic of gender and voice. In other words, our findings are not generalizable to all gender identities. For example, transgender individuals who have undergone voice training have unique experiences with voice and voice training systems [2] that may be different from the experiences of others with different gender identities. We encourage future work to account for the influence of gender identities on experiences with voice and technology, and consider these individuals' experiences in the domain of voice and speech training. More specifically, future work should make design considerations that allow for self-tailoring, including flexible goal-setting [3] and making decisions based on self-assessments rather than external instruction and feedback. We also acknowledge that our findings cannot be generalized to global contexts because the participants of our study were limited to English speakers in the United States.

6 CONCLUSION

While there were no significant effects of user-controlled voice manipulation and the model audio on the self-perceptions of voice and speech, we gained insights into the thought processes, behaviors, and preferences of cisgender and non-binary AFAB participants during speech practice. The participants considered the similarity of their voices and identities with those of the model audio to determine whether or not to gain inspiration from the model audio. The participants also carefully selected which vocal parameters to manipulate and used the feedback of their manipulated voices to make appropriate changes that would help them achieve their ideal voices. Based on these findings, we outline design considerations for a speech practice system that could help users increase their self-perceptions of their voices and speech performances.

Although prior work on speech practice systems relied on external assessments and feedback, we found that self-assessing speech performances allowed the participants to tailor their speech practice to their experiences, needs, and identities. Our findings suggest that gender identity and the experiences tied to that are especially relevant to research that is informed by individuals' self-perceptions. In the context of speech practice systems, future work should consider how voice manipulation and model audio impact self-perceptions, and further develop design considerations based on individuals' experiences that are influenced by their gender identities. We encourage HCI researchers to elevate the experiences and self-perceptions of individuals and advocate for technologies that address the concerns and challenges tied to gender identity.

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